Name: _____

WORKSHEET

# Using exponential models

**1** The exponential model $y = 1000 \times 1.1^x$ is used to describe the value of a \$1000 investment after $x$ years. Complete the table.

| $x$ (years) | 0 | 1 | 2 | 4 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|---|
| $y$ (\$) | | | | | | | |

**2** You invest \$5000 at 6% per annum compounded annually

**a** Create a model for the value of the investment after $n$ years

**b** Complete the table

| $n$ | 1 | 2 | 4 | 6 | 8 | 10 | 12 |
|---|---|---|---|---|---|---|---|
| $f(n)$ | | | | | | | |

**c** How many years does it take to double the initial investment?

**d** Does this doubling time hold for any amount of initial investment?

**3** If Australia's population increases by 1% every year

**a** Use a model to predict the population in the following years

| year | 2013 | 2015 | 2020 | 2030 | 2040 | 2050 | 2100 |
|---|---|---|---|---|---|---|---|
| pop (million) | 23 | | | | | | |

**b** Use trial and error to predict how long (to the nearest 6 months) it will take Australia's population to double at this growth rate.

**4** If Australia's population increases by 2% every year

**a** Predict the population in the following years

| year | 2013 | 2015 | 2020 | 2030 | 2040 | 2050 | 2100 |
|---|---|---|---|---|---|---|---|
| pop (million) | 23 | | | | | | |

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**b** Use trial and error to predict how long (to the nearest 6 months) it will take Australia's population to double at this growth rate.

**5** The value of a computer depreciates 18% every year. If the computer originally cost $1600

**a** Find a model which describes the value of the computer after $x$ years

**b** The value of the computer after 1 year

**c** The value of the computer after 3 years

**d** Estimate how long it takes the value of the computer to drop to $1000

**6** A fish population was found to decrease by 12% for every 1°C increase in water temperature above 17°C. If the initial population is 20 000, find:

**a** a model to describe the fish population at a water temperature of $T$°C .

**b** the fish population in 19°C water.

**c** the fish population in 27°C water.

**d** The fish are considered on the verge of extinction if the population drops below 500. Estimate the water temperature that would drive the fish to the verge of extinction.

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## Answers

**1**

| $x$ (years) | 0 | 1 | 2 | 4 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|---|
| $y$ ($) | 1000 | 1100 | 1210 | 1464.10 | 1771.56 | 2143.59 | 2593.74 |

**2 a** $f(n) = 5000 \times 1.06^n$

**b**

| $n$ | 1 | 2 | 4 | 6 | 8 | 10 | 12 |
|---|---|---|---|---|---|---|---|
| $f(n)$ | 5300 | 5618 | 6312.38 | 7092.60 | 7969.24 | 8954.24 | 10060.98 |

**c** 12 years

**d** yes as long as the interest rate is the same

**3 a**

| year | 2013 | 2015 | 2020 | 2030 | 2040 | 2050 | 2100 |
|---|---|---|---|---|---|---|---|
| pop (million) | 23 | 23.46 | 24.66 | 27.24 | 30.09 | 33.24 | 54.66 |

**b** 70 years

**4 a**

| year | 2013 | 2015 | 2020 | 2030 | 2040 | 2050 | 2100 |
|---|---|---|---|---|---|---|---|
| pop (million) | 23 | 23.93 | 26.42 | 32.21 | 39.26 | 47.89 | 128.81 |

**b** 35.5 years

**5 a** $y = 1600 \times 0.82^x$

**b** $1312

**c** $882.19

**d** 2.37 years

**6 a** $P(T) = 20\,000 \times 0.88^{T-17} \quad T \geq 17$

**b** 15 488

**c** 5570

**d** 46 °C

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